复变函数作业 10.8

1. $\int_0^{3+i} z^2 dz$

(1) 原点至3+i直线段: $z=(3+i)t$, $dz=(3+i)dt$, $t\in(0,1)$ $\int_0^1 (3+i)^2 t^2 dt = \frac{(3+i)^3}{3} = 6+\frac{26}{3}i$

(2) → ↑ : 原式 $= \int_0^3 x^2 dx + i\int_0^1 (3+iy)^2 dy = 9+i(\frac{26}{3}+3i) = 6+\frac{26}{3}i$

(3) ↑ → : 原式 $= i\int_0^1 (iy)^2 dy + \int_0^3 (x+i)^2 dx = -\frac{1}{3} + (6+9i) = 6+\frac{26}{3}i$

2. $\int_0^{3+i} (x^2+iy)\,dz$

(1) $y=x$: $z=(1+i)x$ $dz=(1+i)dx$, $x\in(0,1)$. 原式 $= \int_0^1 (x^2+ix)(1+i)dx = (1+i)\left(\frac{x^3}{3}+\frac{x^2}{2}i\right)\Big|_0^1 = -\frac{1}{6}+\frac{5}{6}i$

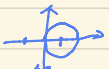
(2) $y=x^2$, $z=x+ix^2$, $dz=(1+2ix)dx$ $x\in(0,1)$ 原式 $= \int_0^1 (x^2+ix)(1+2ix)dx = (1+i)\left(\frac{x^3}{3}+\frac{x^4}{2}i\right)\Big|_0^1 = -\frac{1}{6}+\frac{5}{6}i$

3. $\oint_C \frac{\bar z}{|z|} dz$, C为正向圆周 (1) $|z|=2$ (2) $|z|=4$

$|z|=R$ 时, $z=Re^{i\theta}$, $\bar z = Re^{-i\theta}$ $|z|=R$ 原式 $= \int_0^{2\pi} e^{-i\theta} d(Re^{i\theta}) = \int_0^{2\pi} e^{-i\theta}(iRe^{i\theta})d\theta = iR\int_0^{2\pi} d\theta = 2\pi iR$

∴(1) $R=2$, 原式 $= 4\pi i$ (2) $R=4$, 原式 $= 8\pi i$

7(2) $C: |z-a|=a$ $(a>0)$, 算 $\oint_C \frac{dz}{z^2-a^2}$



$\oint \frac{dz}{z^2-a^2} = \oint \frac{dz}{(z+a)(z-a)}$ 奇点 $z=\pm a$, $z=a$ 不解析, $z=-a$ 解析

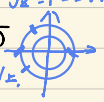
原式 $= \oint_C \frac{\frac{1}{z+a}}{z-a} dz = 2\pi i\left[\frac{1}{z+a}\right]_{z=a} = 2\pi i\cdot\frac{1}{2a} = \frac{\pi i}{a}$

(3) $C: |z-2i|=\frac{3}{2}$, $\oint_C \frac{e^{iz}}{z^2+1} dz$

奇点 $z=\pm i$, 仅 $z=i$ 不解析

原式 $= \oint_C \frac{\frac{e^{iz}}{z+i}}{z-i} dz = 2\pi i\left[\frac{e^{iz}}{z+i}\right]_{z=i} = 2\pi i\cdot\frac{1}{2e} = \frac{\pi}{e}$

(5): $C: |z|=r<1$, $\oint_C \frac{dz}{(z^2-1)(z^2-4)}$



奇点 $z=\pm1$, $e^{\frac{2\pi}{3}i}$, $e^{\frac{4\pi}{3}i}$ 都在 $|z|=1$ 外

在C外全解析, ∴原式 $=0$.

(10) $C: |z|=1$ $\oint_C \frac{e^z}{z^5} dz$

奇点 $z=0$, 在C内. 原式 $= \frac{2\pi i}{4!} f^{(4)}(e^z)\Big|_{z=0} = \frac{\pi i}{24} e^0 = \frac{\pi i}{12}$

8 (1) $\int_{\pi i}^{3\pi i} e^{2z} dz = \left[\frac{e^{2z}}{2}\right]_{-\pi i}^{3\pi i} = \frac{1-1}{2} = 0$

(2) $\int_{\frac{\pi}{8}i}^{0} ch3z\,dz = \left[\frac{sh3z}{3}\right]_{\frac{\pi}{8}i}^{0} = -\frac{1}{3}i$

(3) $\int_{-\pi i}^{\pi i} \sin^2 z\,dz = \left[\frac{z}{2} - \frac{\sin 2z}{4}\right]_{-\pi i}^{\pi i} = \pi i - \frac{1}{2}sh2\pi$

(4) $\int_0^1 z\sin z\,dz = \{z\cos z + \sin z\}_0^1 = \sin 1 - \cos 1$

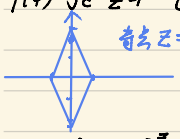
(5) $\int_0^i (z-1)e^{-z} dz = [-ze^{-z}]_0^i = -\sin 1 - i\cos 1$

(6) $\int_1^i \frac{1+\tan z}{\cos^2 z} dz = \left[\frac{(1+\tan z)^2}{2}\right]_1^i = \frac{1}{2}[(1+\tan i)^2 - (1+\tan 1)^2] = \tan i - \tan 1 + \frac{1}{2}\tan^2 i - \frac{1}{2}\tan^2 1$

9(3) $\oint_{C=C_1+C_2} \frac{\cos z}{z^3} dz$, $C_1: |z|=2$ 正 $C_2: |z|=3$ 负

奇点 $z=0$ 在内. $\oint_{C_1} \frac{\cos z}{z^3} = \frac{2\pi i}{2!}(-1) = -\pi i$ $\oint_{C_2} \frac{\cos z}{z^3} = -\left(\frac{2\pi i}{2!}(-1)\right) = \pi i$ ∴ $\oint_C \frac{\cos z}{z^3} = -\pi i + \pi i = 0$

9(4) $\oint_C \frac{dz}{z^4-1}$ C是以 $\pm\frac{1}{2}$, $\pm\frac{1}{2}i$ 为顶点的正向菱形



奇点 $z=i$ 在内部.

原式 $= \frac{2\pi i}{4i}\cdot 1 = \frac{\pi}{2}i$

9(5) $\oint_C \frac{e^z}{(z-a)^2} dz$, $C: |z|=1$ 正 a为 $|a|\neq1$ 的任何复数

①若 $|a|\in(0,1)$, 奇点在C内部. 原式 $= \frac{2\pi i}{2!} e^a = \pi i e^a$.

②若 $|a|>1$, 奇点在C外部, 利用积分定理得原式 $=0$

→ 综上 $\oint_C \frac{e^z}{(z-a)^2} dz = \begin{cases} \pi i e^a & |a|<1 \\ 0 & |a|>1 \end{cases}$

23. 不是.

$u=x+y$, $v=x+y$, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, 但 $\frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x}$. 不满足C-R, 不是关于调和函数.

27. 证明:

① $u=x^2-y^2$ $u_x=2x$ $u_y=-2y$ $u_{xx}=2$. $u_{yy}=-2$ ∴$u_{xx}+u_{yy}=2-2=0$. ∴u在全平面调和

② $v=\frac{y}{x^2+y^2}$ 记 $r^2=x^2+y^2\neq0$. $v_x=-\frac{2xy}{r^6}$ $v_{xx}=-\frac{2y}{r^6}+\frac{8x^2y}{r^8}$ $v_y=\frac{x^2-y^2}{r^6}$ $v_{yy}=-\frac{2y}{r^6}-\frac{4y(x^2-y^2)}{r^8}$

$v_{xx}+v_{yy}=-\frac{4y}{r^6}+\frac{4x^2y+4y^3}{r^8}=0$. v在 $z=0$ 的去线内调和.

③ $u+iv$: $u_x\neq v_y$, C-R方程不成立. 取 $z=1$, 定义域内点, $u_x|_{z=1}=2$, $v_y|_{z=1}=1$. 不等.

$u+iv$ 在定义域内处处不解析

29(1): $u=(x+y)(x^2+4xy+y^2)=x^3+5x^2y+5xy^2+y^3$. $u_x=3x^2+6xy+3y^2=v_y$ $u_y=3x^2+6xy-3y^2=-v_x$

$v=\int(3x^2+6xy-3y^2)dy = 3x^2y+3xy^2-y^3+\varphi(x)$ $v_x=6xy+3y^2+\varphi'(x)$ ∴$\varphi'(x)=-3x^2$ ∴$\varphi(x)=-x^3+C$

∴$v=-x^3+3x^2y+3xy^2-y^3+C$ $f(z)=u+iv=(1+i)z^3+iC$ $C\in\mathbb{R}$

29(3) $u=2(x-1)y=2xy-2y$. $u_x=2y=v_y$ $u_y=2x-2=-v_x$

$v=\int 2y\,dy = y^2+\varphi(x)$ $v_x=\varphi'(x)=2-2x$ ∴$\varphi(x)=2x-x^2+C$.

$f(z)=2(x-1)y+i(y^2-x^2+2x+C)$ $f(2)=i(0^2-2^2+4+C)=-i$ ∴$C=-1$

∴$f(z)=2(x-1)y+i(y^2-x^2+2x-1)$ ∴$f(z)=i(-z^2+2z-1)=-i(z-1)^2$